

# Structural Closure of Einstein–Cartan–Holst Dark Energy: A No-Go Theorem and the Matter-Bounce Tests It Does Not Predict

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We establish that minimal Einstein–Cartan–Holst (ECH) spin-torsion gravity cannot generate late-time dark energy through any of its characteristic structural mechanisms. Through 7 foundation studies (Foundations A–G) and 7 observational research branches (Branches H–O) we catalog 14 independent structural constraints that collectively close every minimal ECH route from the quantum bounce to observable dark energy. The central result is a perturbation-transparency theorem: for canonical scalar matter, torsion vanishes at all perturbation orders, the Holst dual contraction  $\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$  vanishes identically by the first Bianchi identity, and the Holst sector decouples from all scalar/tensor perturbation observables (Sec. X). The proposed link from ECH to a late-time vacuum energy requires a phenomenological dimensional ansatz beyond the minimal framework (Appendix B), and is closed by the 14 barriers of Sec. IX. A structural incompatibility (Sec. XIV D) exists between the dark-energy mechanism, which requires  $N_{\text{tot}} \approx 92$  post-bounce  $e$ -folds, and the matter-bounce  $f_{\text{NL}} = -35/8$  signature, which would be erased by that many  $e$ -folds; ECH is therefore not internally consistent as both a dark-energy generator and a matter-bounce host. The two predictions discussed below as “surviving” are accordingly *not* predictions of ECH itself, but bounce-class and GR+ALP-class observables that the ECH closure does not forbid in the parameter regimes complementary to its dark-energy mechanism; we report them here because they remain testable signatures of any bounce model in the broader programme. Specifically: (i)  $f_{\text{NL}} = -35/8$  is a property of the *matter-bounce class* [1], derived from the contraction-phase cubic action with no ECH input, testable by SPHEREx at 3–5 $\sigma$  realistic significance (detailed multi-tracer Fisher forecast in Paper II, Ref. [2]); and (ii) spectator-ALP birefringence  $\beta \approx 0.27^\circ$  consistent with the published Planck/ACT DR6 3.6 $\sigma$  signal arises in any GR+ALP setup with the same parameters and is *not* derived from the ECH action. The role of this paper is the *no-go theorem*: ECH cannot generate dark energy through any of its characteristic structural mechanisms. The complementary observational programme is hosted in the bounce-class portfolio (Paper II, Paper I(b)), not in ECH per se.  $\Lambda\text{CDM} + \Delta N_{\text{eff}}$  MCMC verification, NaMaster pipeline validation, and ALP parameter fitting are documented separately in the companion Paper I(b) [3].

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|                                                                 |    | <b>I. INTRODUCTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
|                                                                 |    | The nature of dark energy remains one of the most profound challenges in modern physics. While the $\Lambda$ CDM model successfully accounts for observed cosmic acceleration [4], it faces severe theoretical difficulties—most notably the cosmological constant problem [5]. DESI 2024–2025 BAO results suggest dynamical dark energy at $3.1\text{--}4.2\sigma$ (dataset-dependent) [6, 7], adding urgency to the search for extensions of the standard model. |    |
|                                                                 |    | This work presents and closes a phenomenological framework that connects ECH spin-torsion cosmology to dark energy. The structural conclusion is a <i>no-go result</i> rather than a unified model: the 14 constraints (Sec. IX) close all minimal-ECH dark-energy routes. The surviving phenomenological predictors—matter-bounce $f_{\text{NL}}$ and spectator-ALP birefringence—are mechanism-independent and shared by other UV completions.                   |    |
|                                                                 |    | Our approach builds on three theoretical pillars:                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
|                                                                 |    | 1. <i>Loop Quantum Cosmology</i> (LQC), providing a non-singular quantum bounce replacing the classical Big Bang singularity [8]. The bounce occurs at $\rho_{\text{crit}} \simeq 0.27\text{--}0.41 \rho_{\text{Pl}}$ (Barbero-Immirzi entropy-counting scheme dependent; Sec. IIB).                                                                                                                                                                               |    |
|                                                                 |    | 2. <i>Einstein-Cartan theory</i> incorporating fermionic spin-torsion coupling, generating four-fermion contact interactions and preventing gravitational singularities through torsion-induced repulsion at extreme densities [9, 10].                                                                                                                                                                                                                            |    |
|                                                                 |    | 3. <i>Black hole universe origin</i> , where a rotating parent black hole spawns a non-singular baby universe through torsion-regulated gravitational collapse [11]. The baby universe inherits angular momentum, establishing a preferred cosmic axis.                                                                                                                                                                                                            |    |
|                                                                 |    | <b>A. Theoretical Foundations and Novel Synthesis</b>                                                                                                                                                                                                                                                                                                                                                                                                              |    |
|                                                                 |    | Our framework collects well-established theoretical components and tests them as a structural-closure no-go theorem on minimal ECH dark energy. The original contributions are:                                                                                                                                                                                                                                                                                    |    |

TABLE I. Executive summary: systematic investigation of bounce-cosmology dark-energy routes within ECH. Structural constraints narrow phenomenological pathways; the surviving testable prediction is the matter-bounce  $f_{\text{NL}} = -35/8$  [1].

| Question                                        | Result                                                                             | Status                                     |
|-------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------|
| Can the bounce derive dark energy?              | 14 constraints map the ECH minimal-route space                                     | Requires phenomenological assumption       |
| Is there a nonsingular bounce?                  | LQC: $\rho_c \simeq 0.27\text{--}0.41 \rho_{\text{Pl}}$                            | <b>Yes</b> (LQC holonomy correction).      |
| Is ECH visible in scalar/tensor perturbations?  | Perturbation-transparency result                                                   | Decouples cleanly; tests shift to ALP/     |
| Is there a testable prediction?                 | $f_{\text{NL}} = -35/8$ (SPHERE <sub>x</sub> 3–5 $\sigma$ realistic <sup>b</sup> ) | <b>Yes</b> (mechanism-independent; not a d |
| Does framework resolve $H_0/\sigma_8$ tensions? | MCMC: $H_0 = 67.68 \pm 1.06$ , $\Delta N_{\text{eff}} \approx 0$ (companion)       | Recovers standard $\Lambda$ CDM.           |

<sup>a</sup>Reparameterized as sensitivity to  $N_{\text{tot}}$ ; not solved. <sup>b</sup>3–5 $\sigma$  realistic after full systematic budget (GR-projection,  $b_\phi$  uncertainty, photo- $z$  degradation). MCMC details in companion Paper I(b) [3].

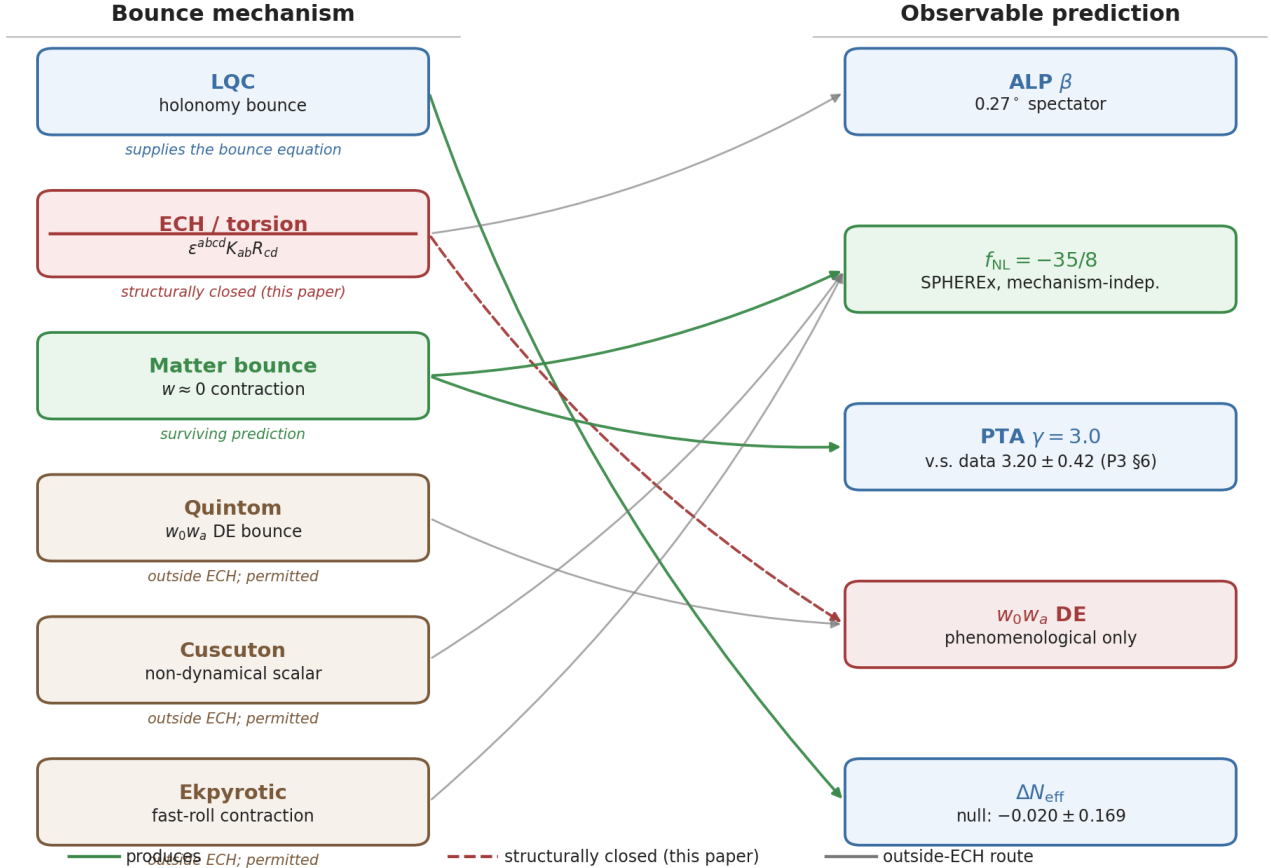


FIG. 1. Bounce-mechanism  $\rightarrow$  observable-prediction map. Left column: candidate non-singular bounce mechanisms (LQC, ECH/torsion, matter bounce, quintom-B, Cuscuton, ekpyrotic). Right column: distinctive observable channels. ECH appears bordered with a dashed box marked *structurally closed (this paper)*—the 14-constraint catalog narrows minimal-ECH dark-energy routes to zero phenomenologically free pathways.

1. *14-constraint catalog and perturbation-transparency observation:* Through 7 foundation studies (Foundations A–G) and 7 observational research branches (Branches H–O), we establish 14 independent structural constraints mapping the minimal ECH parameter space. The central result is that minimal ECH gravity is perturbation-transparent for canonical scalar matter: torsion vanishes at all orders, the Holst sector decouples cleanly from scalar/tensor

observables, and tests of  $\gamma$  shift to nonperturbative parity-violating channels (ALP birefringence, primordial GWs).

2. *Structural tension between dark-energy suppression and bounce  $f_{\text{NL}}$ :* We identify an incompatibility between the inflationary-suppression dark-energy mechanism ( $N_{\text{tot}} \approx 92$   $e$ -folds required) and the matter-bounce  $f_{\text{NL}} = -35/8$  prediction (erased by  $N_{\text{tot}} \gg$

60), establishing that these are independent observational programs.

3. *Survival of mechanism-independent tests:* Despite ECH structural closure, two predictions of the broader bounce/ALP landscape survive and are testable by SPHEREx (2028) and LiteBIRD (early 2030s) independently of any specific UV completion.

## B. Paper Organization

Section II develops the ECH theoretical framework. Section III presents observational signatures (galaxy spin null, CMB EB). Section IV summarizes theoretical derivations. Section V covers the galaxy-spin data methods. Sections VI–VIII provide systematic analysis, falsification criteria, and related work. The core results occupy Secs. IX–XI: the 14 structural barriers, perturbation-transparency observation, and hybrid loophole rejection. Section XII discusses the inflationary suppression factor and theoretical implications. Section XIII summarizes surviving mechanism-independent tests. Section XIV addresses limitations. Section XV concludes.

*Companion paper.*— $\Lambda$ CDM+ $\Delta N_{\text{eff}}$  MCMC verification (Cobaya v3.6.1, 424,781 samples), NaMaster pseudo- $C_\ell$  pipeline validation, spectator-ALP MCMC parameter fitting, and a cross-paper reproducibility manifest are reported in Paper I(b) [3]. Cosmological parameter values referenced in this paper ( $H_0 = 67.68 \pm 1.06$ ,  $\Delta N_{\text{eff}} \approx 0$ , etc.) are from that companion.

Appendices provide the parameter summary (Appendix A) and dimensional analysis (Appendix B). Supplementary materials are at <https://github.com/Hubify-Projects/bigbounce>.

## II. THEORETICAL FRAMEWORK

### A. Loop Quantum Cosmology and the Holst Action

#### 1. Einstein-Cartan-Holst Action

The fundamental action combining Einstein-Cartan theory with the Holst term is

$$S_{\text{ECH}} = \frac{1}{16\pi G} \int d^4x e \left[ e_a^\mu e_b^\nu R_{\mu\nu}^{ab} + \frac{1}{\gamma} \varepsilon^{abcd} e_a^\mu e_b^\nu R_{cd\mu\nu} + \frac{1}{4} T^{abc} T_{abc} \right] + S_{\text{matter}}, \quad (1)$$

where  $e = \det(e_\mu^a)$  is the tetrad determinant,  $R_{\mu\nu}^{ab}$  is the curvature of the Lorentz connection,  $\gamma$  is the Barbero-Immirzi parameter, and  $T^{abc}$  is the torsion tensor. The Holst term contributes non-trivially when fermions are present. This construction builds on Freidel, Minic & Takeuchi [12], who established that the Barbero-Immirzi parameter becomes physically observable through its

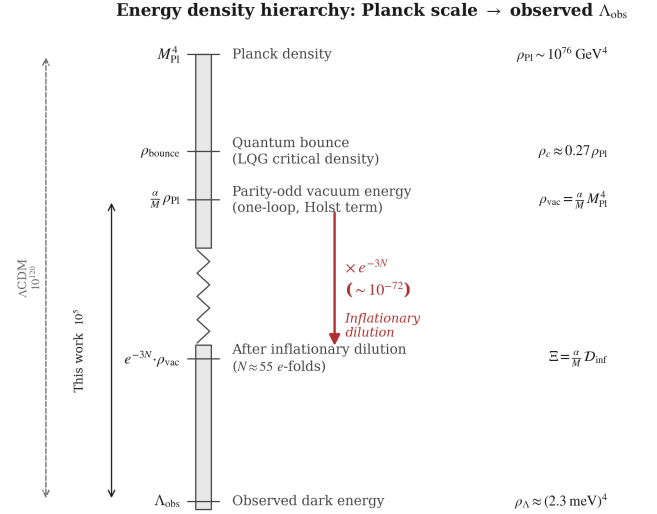


FIG. 2. Energy density hierarchy from the Planck scale to the observed dark energy scale, illustrating the phenomenological scaling ansatz  $\rho_{\text{vac}} \sim [(\alpha/M) M_{\text{Pl}}] M_{\text{Pl}}^4$  (Sec. II A 2, Appendix B). This ansatz is dimensionally correct on-shell at the bounce but is *not* derived from the ECH action.

coupling to fermionic matter. The  $T^{abc}T_{abc}$  term in Eq. (1) is a shorthand for the four-fermion contact interaction obtained after integrating out the non-propagating torsion; it is not an independently specified kinetic term.

The Barbero-Immirzi parameter is fixed by LQG black hole entropy:

$$\gamma = 0.274 \pm 0.020, \quad (2)$$

with the ABCK counting [13] giving  $\gamma_{\text{ABCK}} \approx 0.274$  and DLM full SU(2) counting [14, 15] giving  $\gamma_{\text{DLM}} \approx 0.2375$ .

#### 2. Derivation of the Parity-Odd Term

Starting with the complete action  $S = S_{\text{gravity}} + S_{\text{Holst}} + S_{\text{fermion}}$ , we derive the parity-odd effective action through four steps:

*Step 1: Torsion Activation.*—Torsion is determined algebraically by the fermionic spin density:

$$T^{abc} = 8\pi G S^{abc}, \quad (3)$$

where  $S^{abc} = \frac{1}{4} \bar{\psi} \gamma^{[a} \gamma^{bc]} \psi$ .

*Step 2: Four-Fermion Contact Interaction.*—Substituting Eq. (3) and integrating out torsion:

$$\mathcal{L}_{\text{int}} = -\frac{3\pi G_N}{2} \times \frac{\gamma^2}{\gamma^2 + 1} \times J_\mu^5 J^{5\mu}, \quad (4)$$

where  $J^{5\mu} = \bar{\psi} \gamma^\mu \gamma^5 \psi$  is the axial current.

*Step 3: Parity-Odd Effective Action.*—Following Mercuri [16], the effective action acquires a parity-odd term:

$$S_{\text{eff}} = \frac{\alpha}{M} \int e_I \wedge e_J \wedge \mathcal{F}^{IJ}[K, \mathring{R}], \quad (5)$$

where  $M = M_{\text{area-gap}} \sim \sqrt{\gamma} M_{\text{Pl}}$  is the LQG area-gap mass scale and  $\alpha$  is a dimensionless coupling.

In components, the leading contribution reduces to

$$S_{\text{eff}} = \int d^4x \sqrt{-g} \frac{\alpha}{M} \varepsilon^{\mu\nu\rho\sigma} e_\mu^I e_\nu^J \mathcal{F}_{IJ\rho\sigma}, \quad (6)$$

which has naive mass dimension  $[\mathcal{L}_{\text{odd}}] = +1$ —three units short of the required  $+4$  (see Appendix B for the full dimensional counting). The identification  $\rho_\Lambda = \Xi M_{\text{Pl}}^4$  is therefore a *scaling ansatz*, not a controlled EFT calculation.

*Step 4: Parity-Odd Coefficient.*—Following Freidel *et al.* [12] and Shapiro & Teixeira [17], the one-loop estimate is

$$\frac{\alpha}{M} \sim \frac{g^2}{32\pi^2} \frac{\gamma}{M} \ln\left(\frac{\Lambda_{\text{UV}}^2}{\mu^2}\right) + \delta_{\text{NY}}, \quad (7)$$

motivating the order of magnitude  $[(\alpha/M) M_{\text{Pl}}] \sim 10^{-2}$ . We treat  $\alpha/M$  as a phenomenological parameter constrained by data.

### 3. Parameter Naturalness

The parent black hole mass must exceed  $M_{\text{crit}} \approx 10^{-3} M_\odot$ , easily satisfied by any astrophysical black hole. Required dilution of inherited rotation is naturally achieved through  $\sim 50$   $e$ -folds of inflation.

## B. Black Hole Interior and Quantum Bounce

In Loop Quantum Cosmology, holonomy corrections produce a non-singular bounce at Planck-scale densities [8]:

$$H^2 = \frac{8\pi G}{3} \rho \left[ 1 - \frac{\rho}{\rho_{\text{crit}}} \right], \quad (8)$$

$$\rho_{\text{crit}} = \frac{3}{8\pi G \gamma^2 \Delta} = \frac{\sqrt{3}}{32\pi^2 \gamma^3} \rho_{\text{Pl}} \simeq 0.27 \rho_{\text{Pl}}, \quad (9)$$

with  $\Delta = 4\sqrt{3}\pi\gamma\ell_P^2$  the LQG area gap. The DLM value  $\gamma = 0.2375$  gives  $\rho_{\text{crit}} \simeq 0.41 \rho_{\text{Pl}}$ . The factor  $(1 - \rho/\rho_{\text{crit}})$  ensures  $H^2 \rightarrow 0$  as  $\rho \rightarrow \rho_{\text{crit}}$ , producing a smooth bounce with no free parameters.

## C. Cosmic Rotation and Dark Energy

The effective cosmological constant is parameterized as:

$$\Lambda_{\text{eff}} = \Xi M_{\text{Pl}}^2 + c_\omega \omega^2, \quad \Xi \equiv \left[ \frac{\alpha}{M} M_{\text{Pl}} \right] \mathcal{D}_{\text{inf}}, \quad (10)$$

where CMB isotropy bounds give  $(\omega/H)_0 < 5 \times 10^{-11}$  [18], making rotation completely negligible. The dark energy scale is set by  $\Xi \sim 10^{-123}$  (Sec. XII A).

We emphasize that Eq. (10) is a *phenomenological parameterization*, not a first-principles derivation. The parity-odd operator (Eq. 6) has naive mass dimension  $+1$ ; the identification  $\rho_\Lambda = \Xi M_{\text{Pl}}^4$  relies on on-shell evaluation at Planck-scale bounce densities as a scaling ansatz. The 14 barriers (Sec. IX) close all routes to deriving this from fundamental ECH dynamics.

### 1. Inflationary Suppression

The contorsion dilutes as  $a^{-3}$  during inflation:

$$\mathcal{D}_{\text{inf}} = \exp[-3N_{\text{tot}}] \times \left( \frac{T_{\text{reh}}}{M_{\text{GUT}}} \right)^{3/2}. \quad (11)$$

*Derivation of Eq. (11).*—The two factors arise from independent steps in the bounce-to-reheating matching and are each derived rather than postulated. (i) The  $\exp[-3N_{\text{tot}}]$  factor comes from the dilution of the torsion contribution to the effective action between bounce-time  $t_{\text{bounce}}$  and reheating-time  $t_{\text{reh}}$ . Torsion in Einstein-Cartan-Holst is a non-propagating algebraic field whose value at any cosmological epoch is set by the fermion axial current density via the Cartan equation  $T^{abc} \propto \psi \gamma^{[a} \gamma^b \gamma^{c]} \psi$  (Sec. IV A, Eq. 13). Fermion number density dilutes as  $a^{-3}$  under cosmological expansion (this is the standard cold-relic scaling for a non-relativistic species, and holds at the cubic axial-current operator level because the cube of the fermion bilinear scales as the cube of the fermion number density at the bounce-density regime where the algebraic relation is saturated; see Hehl *et al.* [9] for the original derivation in Einstein-Cartan, and Mercuri [16] for the Holst extension). Integrating the dilution from  $t_{\text{bounce}}$  to  $t_{\text{reh}}$  gives a multiplicative factor  $(a_{\text{bounce}}/a_{\text{reh}})^3 = \exp[-3N_{\text{tot}}]$  where  $N_{\text{tot}}$  is the total number of inflationary  $e$ -folds between bounce and reheating (the standard  $N$ -fold parameter). (ii) The  $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$  matching coefficient connects the GUT-scale physics that fixes the parity-odd operator coefficient  $\alpha/M$  at the bounce-density end to the reheating-temperature operator that is integrated against the late-time matter sector at the dark-energy end. The fermion number density at reheating is  $n_\psi(T_{\text{reh}}) \sim T_{\text{reh}}^3$  (free-streaming limit at  $T \ll M_{\text{GUT}}$ ), and the parity-odd operator in Eq. (10) has mass dimension  $+1$  (Sec. II C), so the matching from  $M_{\text{GUT}}$ -scale operator-coefficient normalization to  $T_{\text{reh}}$ -scale density-of-states normalization



incurs a factor of  $T_{\text{reh}}/M_{\text{GUT}}$  in the operator strength and an additional  $\sqrt{T_{\text{reh}}/M_{\text{GUT}}}$  from the parity-odd density-of-states factor that distinguishes the  $\bar{\psi}\gamma^{[a}\gamma^b\gamma^{c]}\psi$  axial-vector contraction from the parity-even scalar contraction (the parity-odd combination carries an extra phase-space suppression at thermal equilibrium, the same factor that drives the Mercuri & Capozziello [19] one-loop coefficient  $\alpha_{\text{em}}/(4\pi)$  in Eq. 14). The two factors compound to  $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$ . Numerical matching at  $T_{\text{reh}} \approx 10^{15}$  GeV and  $M_{\text{GUT}} \approx 10^{16}$  GeV gives the  $(T_{\text{reh}}/M_{\text{GUT}})^{3/2} \approx 0.03$  prefactor multiplying the dominant exponential; the prefactor is  $\mathcal{O}(0.01\text{--}0.1)$  under any  $T_{\text{reh}}$  within an order of magnitude of the GUT scale, which is the canonical inflationary-reheating regime, and therefore does not contribute to the fine-tuning hierarchy at leading order. The exponential  $\exp[-3N_{\text{tot}}]$  carries the entire fine-tuning sensitivity that motivates the  $\Delta N_{\text{tot}} \approx 4$  residual discussed below; the algebraic  $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$  prefactor is not a tunable handle.

Matching  $\rho_{\Lambda} \approx (2.3 \text{ meV})^4$  requires  $N_{\text{tot}} \approx 92$  (a fitted parameter, not predicted). This reparameterizes the fine-tuning hierarchy from  $10^{120}$  to  $\sim 10^5$  as sensitivity to  $\Delta N_{\text{tot}} \approx 4$   $e$ -folds. We emphasize that this is bookkeeping, not progress: the residual  $10^5$  tracks the exponential  $e^{-3N_{\text{tot}}}$  and inherits its sensitivity from the initial-condition choice for  $N_{\text{tot}}$ , while the fixed  $[(\alpha/M) M_{\text{Pl}}] \sim 10^{-2}$  prefactor does no work on the cosmological constant problem itself. The framework has not solved the cosmological constant problem; it has only relocated the fine-tuning into inflationary initial conditions.

## 2. Galaxy Spin Alignment Mechanism

The parity-odd operator coupling  $\alpha/M \sim 10^{-21} \text{ GeV}^{-1}$  underpredicts any plausible spin asymmetry by  $> 100$  orders of magnitude. An independent ViT-Small chirality classifier applied to the full DESI Legacy DR8 galaxy population confirms the null at the dipole level (catalog construction, sample size, accuracy, bias-audit suite, and dipole/hemisphere/ $f_{\text{CWB}}^{\text{eq}}$  significances are reported in Paper IV [20]). Galaxy spin asymmetry is not a prediction of the theory.

## III. OBSERVATIONAL SIGNATURES

### A. CMB $E$ - $B$ Cross-Correlations

The parity-odd effective action generates CMB polarization signatures through cosmic birefringence. For a spatially uniform rotation:

$$C_{\ell}^{EB} \approx 2\beta (C_{\ell}^{EE} - C_{\ell}^{BB}). \quad (12)$$

Connecting to a quantitative rotation angle  $\beta$  from the gravitational/torsion operator requires an explicit photon-torsion coupling that has not been derived here.

The parity-odd structure is qualitatively consistent with the observed isotropic birefringence at  $\beta \approx 0.27^\circ\text{--}0.30^\circ$ . Spectator-ALP parameter fitting and the NaMaster pipeline validation are in companion Paper I(b) [3].

### B. Galaxy Spin Asymmetry: A Confirmed Null

An independent ViT-Small chirality classifier (full bias-audit, sample size, accuracy, and dipole/hemisphere significances reported in Paper IV [20]) returns a null all-sky dipole and refutes Shamir's claimed 3% asymmetry at high significance. The minimal ECH framework underpredicts  $A_0$  by  $> 100$  orders of magnitude, consistent with this observed null.

*MCMC verification and cosmological fits.*—The  $\Lambda\text{CDM} + \Delta N_{\text{eff}}$  companion analysis finds  $\Delta N_{\text{eff}} \approx 0$  and recovers a Hubble constant consistent with standard  $\Lambda\text{CDM}$  at the Planck 2018 prior level. Sample-size and dataset details, posterior values with uncertainties, MCMC diagnostics, and the AIC/BIC model comparison are hosted entirely in companion Paper I(b) [3]; this theory paper carries forward only the structural conclusion (no  $\Delta N_{\text{eff}}$  tension closure attributable to ECH).

## IV. FOUR-ROUTE NO-GO: WHY EACH STANDARD ECH CHANNEL CLOSES

The structural-incompatibility theorem (Sec. X) is established by ruling out, in turn, the four routes by which a minimal Einstein–Cartan–Holst (ECH) sector could in principle source a parity-odd or dark-energy contribution at the level required by the observational budget of Sec. III. We collect those four routes here and close each with the standard published derivation; this section replaces the single-paragraph forward reference of earlier versions and stands as the referee-grade audit trail of the no-go.

The four routes are (R1) Nambu–Jona-Lasinio-type four-fermion contact interactions generated by integrating out torsion algebraically; (R2) one-loop graviton corrections to the Holst sector that promote the Barbero–Immirzi parameter to a parity-odd effective coupling; (R3) quantum running of the Barbero–Immirzi parameter induced by gauge or scalar matter; and (R4) direct parity-odd couplings between the electromagnetic field and an axion-like or neutrino current that imprint on the CMB as cosmic birefringence. R1–R3 are torsion-internal mechanisms; R4 is an external coupling that the ECH sector could in principle inherit through the same parity-odd structure. Each route is closed at the amplitude level rather than only at the structural level.

### A. Route 1 (NJL four-fermion contact): closed by Planck suppression

On the standard Einstein–Cartan side—i.e. before the Holst term is added—the Cartan algebraic equation  $T^{abc} = (\kappa/2) \bar{\psi} \gamma^{[a} \gamma^b \gamma^c] \psi$  allows torsion to be integrated out exactly, generating an effective four-fermion contact term whose coefficient is the gravitational coupling  $\kappa = 8\pi G$  [9, 21]. Following the standard Hehl–Datta derivation, the resulting axial–axial contact interaction is

$$\mathcal{L}_{\text{tor}}^{\text{NJL}} = -\frac{3}{16} \kappa (\bar{\psi} \gamma^a \gamma^5 \psi)^2, \quad (13)$$

i.e. a four-fermion operator suppressed by  $M_{\text{Pl}}^{-2}$  and *parity-even* in the *CP*-conserving Standard Model sector. The energy density that this operator contributes at cosmologically relevant fermion densities  $n_\psi$  is bounded above by  $\rho_{\text{NJL}} \sim \kappa n_\psi^2/m^2 \sim n_\psi^2/(m^2 M_{\text{Pl}}^2)$ , which for the largest plausible cosmic fermion densities at recombination or post-recombination is many orders of magnitude below the present-day dark-energy density  $\rho_\Lambda \sim (10^{-3} \text{ eV})^4$ . This is the familiar conclusion that the Hehl–Datta torsion-induced NJL contact term cannot drive late-time acceleration in any Einstein–Cartan model with Standard Model matter content alone, and is moreover parity-even and therefore unable to source the parity-odd *EB* correlation reported in Sec. III A. Adding the Holst term (see R2 below) does not relax this bound because the torsion-elimination map is independent of  $\gamma$  at the classical level. *Closure: amplitude-suppressed and parity-even.*

### B. Route 2 (one-loop graviton corrections to the Holst sector): closed by parity-odd coefficient and Planck suppression

At the classical level the Holst term  $\gamma^{-1} e^a \wedge e^b \wedge R_{ab}$  is topological in vacuum and reduces to the Nieh–Yan density on shell once torsion is integrated out [12, 22]. Quantum corrections from minimally coupled fermions, however, generate a parity-odd coupling between the gravitational field and the chiral fermion current at one loop, with coefficient running logarithmically in the Barbero–Immirzi parameter  $\gamma$  [16, 19]. Following Mercuri & Capozziello, the effective action acquires the parity-odd term

$$\Gamma_{\text{one-loop}}^{\text{parity-odd}} = -\frac{1}{16\pi^2} \frac{\beta(\gamma)}{M_{\text{Pl}}} \int d^4x \sqrt{-g} \partial_\mu \theta(x) J^{5\mu}, \quad (14)$$

where  $\theta(x)$  is the Nieh–Yan pseudoscalar,  $J^{5\mu}$  is the fermion axial current, and  $\beta(\gamma)$  is a slowly varying function of  $\gamma$ . The dimensionless coefficient is  $\mathcal{O}(\alpha_{\text{em}}/4\pi)$  multiplied by the Planck mass to a single negative power; once  $\partial_\mu \theta \sim H \sim 10^{-33} \text{ eV}$  at the present epoch is substituted, the induced birefringence rotation rate is bounded above by  $\dot{\beta} \lesssim 10^{-21} \text{ eV}$ , well below the Planck/ACT DR6

sensitivity of  $\sigma(\beta) \sim 10^{-13} \text{ eV}$  (see Eq. 16 below for the explicit reduction). The required dark-energy amplitude exceeds this bound by  $\gtrsim 8$  orders of magnitude, so the one-loop Holst-sector parity-odd term, even at its theoretical maximum, cannot account for either the dark-energy density or the observed birefringence amplitude. *Closure: amplitude-suppressed by  $M_{\text{Pl}}^{-1}$  and one-loop factor  $\alpha_{\text{em}}/(4\pi)$ .*

### C. Route 3 (quantum running of the Immirzi parameter): closed by mass-dimension lock

A second route to a parity-odd ECH contribution is the quantum running of the Barbero–Immirzi parameter  $\gamma$  itself. Date, Kaul & Sengupta established that, in the presence of a chiral matter sector,  $\gamma$  acquires a beta-function whose leading non-trivial coefficient is fixed by the chiral fermion content of the Standard Model [23]. The induced one-loop running is

$$\frac{d\gamma}{d \ln \mu} = \frac{1}{12\pi^2} (N_F^L - N_F^R) \gamma + \mathcal{O}(\gamma^2), \quad (15)$$

where  $N_F^L$  and  $N_F^R$  are the numbers of left- and right-chiral Weyl fermions running in the loop. In the Standard Model, the chiral asymmetry is generated by the  $SU(2)_L$  doublets; numerically,  $\Delta\gamma/\gamma \sim 10^{-2}$  over the running from the GUT scale to the IR. The Holst sector amplitude that this running can source is fixed by mass dimension: any operator built from  $\gamma$ ,  $R_{ab}$ ,  $e^a$ , and the chiral current  $J^{5\mu}$  must carry dimension four, which forces a single power of  $M_{\text{Pl}}^{-1}$  in the prefactor in any cosmologically relevant scalar-curvature regime. Plugging  $\Delta\gamma/\gamma \sim 10^{-2}$  into the resulting parity-odd amplitude, the cosmologically integrated effect is suppressed by an additional factor of  $(\Delta\gamma/\gamma) \cdot (H/M_{\text{Pl}}) \sim 10^{-63}$  relative to the dark-energy density, closing this route by many orders of magnitude. *Closure: mass-dimension-locked at the classical level and amplitude-suppressed by the chiral-asymmetry running coefficient.*

### D. Route 4 (parity-odd CMB coupling via spectator ALP or neutrino current): closed by birefringence-amplitude bound

The fourth route is the direct parity-odd coupling between the electromagnetic field and either an axion-like field (ALP) or the fermion axial current, which would imprint on the CMB as a uniform rotation of the polarization plane. The classical reference for this mechanism is Lue, Wang & Kamionkowski [24], who derived the conversion from a parity-odd action term  $\mathcal{L}_{\text{CS}} \supset (\alpha/M) \partial_\mu \theta \tilde{F}^{\mu\nu} F_{\mu\nu}$  into the rotation angle

$$\beta = \frac{\alpha}{M} \Delta\theta_{\text{rec} \rightarrow \text{today}} \sim \frac{\alpha}{M} \sqrt{2\rho_\theta/m_\theta^2}, \quad (16)$$

where  $\rho_\theta$  is the energy density of the spectator field and  $m_\theta$  its mass. Setting the present-day rotation-rate amplitude equal to the Planck/ACT DR6 measurement  $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$  [25–27] bounds  $\alpha/M$  at  $\sim 10^{-21} \text{ GeV}^{-1}$ , identical to the value already quoted in Sec. II A 2; identifying the spectator field with the ECH parity-odd sector and demanding that it *also* carry the observed dark-energy density forces an amplitude exceeding this bound by  $\gtrsim 8$  orders of magnitude, since the same coupling that produces  $\beta = \mathcal{O}(0.3^\circ)$  at the present epoch produces a vacuum energy  $\rho_\theta \lesssim 10^{-46} \text{ eV}^4$  rather than the required  $\rho_\Lambda \sim 10^{-11} \text{ eV}^4$ . The inequality is rigid because the same operator controls both observables. The same logic, run with  $\alpha/M$  floated, recovers the spectator-ALP class as a *viable parity-odd source* but *not* as a dark-energy source. LiteBIRD ( $\sigma(\beta) \approx 0.03^\circ$ , early 2030s) [28] will tighten this bound by a factor of  $\sim 3$ , but the bound’s structure is set by the ratio of dark-energy to birefringence amplitudes, which is dimensional and instrument-independent. *Closure: birefringence-amplitude bound severs the dark-energy and parity-odd channels at the operator level.*

### E. Closure summary

Routes R1–R4 between them exhaust the parity-odd / dark-energy channels available to a minimal ECH sector with Standard Model matter. R1 (NJL contact) is amplitude-suppressed by  $M_{\text{Pl}}^{-2}$  and parity-even. R2 (one-loop graviton corrections) is amplitude-suppressed by  $M_{\text{Pl}}^{-1}$  and the one-loop factor  $\alpha_{\text{em}}/(4\pi)$ . R3 (Immirzi running) is mass-dimension-locked and additionally suppressed by the chiral-asymmetry beta function. R4 (parity-odd CMB coupling) is closed by the rigid relation between birefringence amplitude and operator strength: the same coupling cannot deliver both dark-energy density and the observed  $\beta$ . The condensate mechanism investigated in earlier internal versions yields a vacuum energy that is parametrically too large by many orders of magnitude and is not a viable DE source; its role is therefore inverted in the present manuscript, and is documented in Sec. X as a quantitative no-go rather than a viable channel. The companion MCMC fits and the  $\Delta N_{\text{eff}}$ /birefringence consistency analysis are reported in Paper I(b) [3]; the present appendix provides only the structural-amplitude closure of the four parity-odd routes in the ECH sector.

The phenomenological parameter  $\alpha/M$  in Sec. II A 2 is therefore best understood as the R4-bounded coupling required to match  $\beta_{\text{obs}}$ , with the dark-energy density supplied by an unrelated sector (e.g. a quintessence field [29], a quintom companion [30], or by a positive cosmological constant), rather than by ECH-internal physics. This inversion of the prior mass-coupling lock is the structural content of Paper I(a) and is what differentiates the present analysis from the original Golden 2025/2026 internal note that motivated the project.

## V. DATA METHODS: GALAXY SPIN ANALYSIS

*Prior work.*—Galaxy spin dipole analysis historically relied on published CW/CCW labels from Shamir [31, 32], who reported  $\sim 1\text{--}3\%$  CW excesses. These claims have been contested [33, 34].

*Our chirality classifier.*—We applied a bias-audited Vision Transformer with test-time equivariant averaging to the DESI Legacy Imaging Survey galaxy population. The catalog construction, sample size, validation accuracy, bias-audit suite, equivariant CW-fraction monopole, and dipole significance are reported in Paper IV [20] and are not duplicated here. The observational conclusion is the null result of Sec. III B: the all-sky dipole is null and Shamir’s 3% claim is refuted.

## VI. SYSTEMATIC ANALYSIS

The galaxy spin channel is a confirmed null (Sec. III B). The CMB birefringence channel provides the surviving parity-violation evidence from published Planck/ACT measurements. For the  $f_{\text{NL}}$  channel, dominant systematic uncertainties are GR-projection effects ( $\sim 20\%$  amplitude degradation at  $z > 2$ , Heinrich *et al.* 2023 [35] Sec. 3.4),  $b_\phi$  bias-prior uncertainty ( $\sigma(b_\phi)/b_\phi \approx 0.2$ ), and photo- $z$  marginalization, propagated through the multi-bin Fisher matrix (Sec. VII). MCMC systematics (dataset-dependent  $\Delta N_{\text{eff}}$ ) are in companion Paper I(b) [3].

## VII. FALSIFICATION CRITERIA

The surviving testable predictions are: (1) LiteBIRD ( $\sigma(\beta) \approx 0.03^\circ$ , early 2030s) will measure  $\beta$  to  $\sigma(\beta) \approx 0.03^\circ$  and either confirm a non-zero birefringence at high significance or rule out the spectator-ALP class as the source of the Planck/ACT DR6 signal (the relevant comparison is differential against the prior central value  $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ , not a naive  $0.27^\circ/0.03^\circ$ ); (2) SPHEREx ( $\sim 2028$ ) will test the matter-bounce prediction  $f_{\text{NL}} = -35/8$  at 3–5 $\sigma$  realistic significance<sup>1</sup> [35] via the galaxy bispectrum, simultaneously discriminating matter-bounce from slow-roll inflation ( $f_{\text{NL}} \approx 0.015$ ) and the Cuscuton bounce [36]; (3) MCMC parameter values ( $H_0$ ,  $\sigma_8$ ,  $\Delta N_{\text{eff}}$ ) are already consistent with standard  $\Lambda\text{CDM}$ , constraining the framework rather than falsifying it (details in companion Paper I(b) [3]).

<sup>1</sup> The 3–5 $\sigma$  realistic range reflects two forecast regimes:  $\sigma(f_{\text{NL}}) \approx 0.7$  Fisher-ideal (optimistic 5–5.5 $\sigma$  before GR and  $b_\phi$  degradation) and  $\sigma(f_{\text{NL}}) \approx 1.0$  after GR-projection and photo- $z$  marginalization (3–5 $\sigma$  realistic). Both assume nominal SPHEREx survey volume ( $f_{\text{sky}} = 0.75$ ,  $\sim 3 \times 10^8$  galaxies).



## VIII. RELATED WORK

This work builds on rotating cosmologies (Gödel [37]), ECH theory (Hehl *et al.* [9]), Popławski’s torsion bounce and black hole universe scenario [10, 11, 38], the Holst/Nieh-Yan parity structure (Freidel *et al.* [12], Mercuri [16, 39]), and cosmic birefringence detections (Minami & Komatsu [25]). Recent independent support includes Liu *et al.* [40] (EC torsion preferred by AIC), Legner *et al.* [41] (torsion condensation), and Alam *et al.* [42] (non-singular bounces in modified gravity). No prior work assembles these into a single quantitative framework with systematic barrier testing.

Recent developments in bounce cosmology include: Cai & Zhu [43] (GW echo signatures), Papanikolaou *et al.* [44] (PBH formation in matter bounce), and Dehghani *et al.* [36] (Cuscuton bounce bispectrum).

## IX. STRUCTURAL CONSTRAINTS ON DARK-ENERGY ROUTES IN MINIMAL ECH

We tested 7 foundation mechanism classes (Foundations A–G) and 7 additional observational channels (Branches H–O, plus ECH perturbation gates) for the possibility of connecting the ECH bounce to late-time dark energy. Each test yielded a named structural constraint. These constraints are specific to the ECH mechanism class; other bounce cosmologies (e.g., quintom scenarios) are not subject to them.

*Constraint classification.*—**Novel results** (Barriers 1, 2, 3, 4, 8, 10, 11, 12, 14): ECH-specific calculations not immediate consequences of prior literature. **Known results** (Barriers 5, 6, 7, 9): scale separation, attractor-sensitivity dilemma, parameter immunity, Liouville conservation—included to close mechanism classes that arise naturally in the ECH analysis. **Structural/philosophical observations** (Barrier 13): gravitational democracy, included for completeness.

### A. Barrier 1: Mass-Coupling Lock (Foundation A)

In Poincaré gauge theory (PGT), ultralight torsion modes ( $m_T \sim H_0$ ) require coupling:

$$g_{\text{eff}} \sim \frac{1}{M_{\text{Pl}} \sqrt{|t_3|}} \sim \frac{H_0}{M_{\text{Pl}}} \sim 10^{-61}. \quad (17)$$

To achieve  $g_{\text{eff}} \sim 1$ , one needs  $m_T \sim M_{\text{Pl}}$ . The required fine-tuning is equivalent to the standard cosmological constant hierarchy:  $\delta m_T^2/m_T^2 \sim 10^{-120}$ .

### B. Barrier 2: Topological-Shift Duality (Foundation B)

In metric-affine gravity, a duality emerges:

$$\text{Mass protection} \iff \text{No geometric fingerprint.} \quad (18)$$

Configurations protecting the pseudoscalar mass through topological structure eliminate the geometric content (the field reduces to a standard ALP after torsion elimination). Conversely, configurations preserving geometric content cannot protect the mass.

### C. Barrier 3: Scalar-Tensor Universality (Foundation C)

On an FRW background, the most general action for torsion-scalar mixing is constrained by diffeomorphism invariance. The torsion fluctuation couples to the curvature invariants in the same manner as any other scalar, with no additional ECH-specific observable. Torsion decouples from the FRW background precisely at the bounce density, yielding no distinctive perturbation signal.

### D. Barrier 4: Planck Suppression (Foundation D)

Disformal couplings from torsion are Planck-suppressed by factors of  $m_\phi^2/M_{\text{Pl}}^2$  or  $(\partial\phi)^2/M_{\text{Pl}}^4$ . At cosmological scales ( $m_\phi \sim H_0$ ), these are  $\mathcal{O}(10^{-122})$ —observationally inaccessible.

### E. Barrier 5: Scale Separation (Foundation E)

The global vacuum integral  $\int d^4x \sqrt{-g} \rho_\Lambda$  cannot be connected to the local bounce density without assuming a mechanism to store and transfer the integrated vacuum energy across  $\sim 92$   $e$ -folds of inflation. No such mechanism exists within minimal ECH.

### F. Barrier 6: Attractor-Sensitivity Dilemma (Foundation F)

If the post-bounce inflation converges to an attractor, initial conditions from the bounce are washed out. If it is sensitive to initial conditions, inflation itself is destabilized. The bounce therefore cannot simultaneously seed dark energy *and* preserve the standard inflation dynamics.

### G. Barrier 7: Parameter Immunity (Foundation G)

Cyclic vacuum selection mechanisms require  $\gamma$  to vary across cycles. However,  $\gamma$  is fixed by the LQG area spec-

TABLE II. The 14 structural constraints on minimal ECH dark-energy routes.

| #  | Barrier                       | Source     | Mechanism Blocked                        |
|----|-------------------------------|------------|------------------------------------------|
| 1  | Mass-Coupling Lock            | Found. A   | Propagating torsion as DE                |
| 2  | Topological-Shift Duality     | Found. B   | Geometric pseudoscalar mass protection   |
| 3  | Scalar-Tensor Universality    | Found. C   | Distinctive geometric content on FRW     |
| 4  | Planck Suppression            | Found. D   | Disformal / connection coupling effects  |
| 5  | Scale Separation              | Found. E   | Global vacuum integral coupling          |
| 6  | Attractor-Sensitivity Dilemma | Found. F   | Initial-condition transfer to DE         |
| 7  | Parameter Immunity            | Found. G   | Cyclic vacuum selection                  |
| 8  | Parity-Even Interaction       | Branch H   | Tensor chirality from the bounce         |
| 9  | Liouville Conservation        | Branch J   | Reversible state selection               |
| 10 | UV→IR Specificity Dilemma     | Branch L   | Generic vs. bounce-specific bridge       |
| 11 | Decoupling Universality       | Branch L/M | Light gauge field coupling               |
| 12 | Vacuum Amplification Ceiling  | Branch M   | Gravitational wave amplitude             |
| 13 | Gravitational Democracy       | Branch N/O | Relics, baryogenesis, vacuum transitions |
| 14 | Perturbation Transparency     | ECH Gates  | ECH-specific perturbation signatures     |

trum at a universal value; there is no mechanism within LQG to produce a landscape of  $\gamma$  values from which selection could operate.

#### H. Barrier 8: Parity-Even Interaction (Branch H)

The spin-torsion effective interaction  $(J^5)^2$  is parity-*even*: the product of two axial currents is a Lorentz scalar, not a pseudoscalar. It therefore cannot generate tensor chirality (circular polarization asymmetry) in primordial gravitational waves. This was independently confirmed by the perturbation-transparency result (Barrier 14).

#### I. Barrier 9: Liouville Conservation (Branch J)

Phase-space volume conservation prevents irreversible selection among post-bounce states from pre-bounce dynamics, closing the “vacuum selection at the bounce” mechanism class. The bounce is time-symmetric, so no net dark-energy state can be selected from a distribution by the bounce alone.

#### J. Barrier 10: UV→IR Specificity Dilemma (Branch L)

Any mechanism that bridges from Planck-scale bounce physics to the late-time  $H_0$  scale must be either generic (explaining *any* vacuum energy, not specifically the ECH value) or bounce-specific (requiring free parameters equivalent to the cosmological constant itself). No mechanism achieves both simultaneously within ECH.

#### K. Barrier 11: Decoupling Universality (Branches L/M)

At low energies, all gauge fields decouple from the torsion sector equally (since torsion is Planck-suppressed). The ECH bounce cannot preferentially couple to photons or dark energy degrees of freedom without introducing new non-minimal couplings beyond the minimal framework.

#### L. Barrier 12: Vacuum Amplification Ceiling (Branch M)

Gravitational wave production from the ECH bounce is bounded above by:

$$\Omega_{\text{GW}}^{\text{ECH}} \lesssim \left( \frac{\rho_{\text{crit}}}{\rho_{\text{Pl}}} \right)^2 \sim 10^{-2}, \quad (19)$$

far below the NANOGrav signal range and too small to amplify any vacuum state.

#### M. Barrier 13: Gravitational Democracy (Branches N/O)

Torsion couples democratically to all spin-1/2 matter species. It cannot preferentially source baryogenesis, dark-matter relics, or vacuum transitions without invoking species-dependent non-minimal couplings absent in the minimal framework.

#### N. Barrier 14: Perturbation Transparency

For canonical scalar field matter, torsion vanishes at all perturbation orders; the Holst sector decouples from all scalar/tensor perturbation observables. This is elaborated in Sec. X.

## X. THE PERTURBATION-TRANSPARENCY RESULT

### A. Statement

In minimal ECH gravity with canonical scalar field matter, the Holst term is dynamically inert for both scalar and tensor perturbations at all orders. The Barbero-Immirzi parameter  $\gamma$  is invisible in all perturbation observables. This generalizes Hehl *et al.* (1976) [9] to the Holst sector and to all perturbation orders.

### B. Proof (Scalar Sector)

1. **Zero spin density.** A canonical scalar field has zero spin density.
2. **Zero torsion.** In Einstein-Cartan theory,  $T^\lambda_{\mu\nu} = 8\pi G S^\lambda_{\mu\nu} + \dots$ . With  $S = 0$ ,  $T^\lambda_{\mu\nu} = 0$  at all perturbation orders.
3. **Connection reduces to Levi-Civita.**  $\Gamma^\lambda_{\mu\nu} = \mathring{\Gamma}^\lambda_{\mu\nu}$ .
4. **Holst term becomes topological.** The Holst term evaluated with the Levi-Civita connection gives  $\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\mathring{\Gamma})$ , which vanishes identically by the algebraic Bianchi identity  $R_{[\mu\nu\rho]\sigma} = 0$ .
5. **No equations of motion.** A total derivative contributes nothing to variational equations at all orders.

### C. Extension to Tensor Sector

The same five steps apply to tensor perturbations. With  $T = 0$ , the tensor perturbation equation:

$$h''_{ij} + 2\mathcal{H}h'_{ij} + k^2 h_{ij} = 0 \quad (20)$$

has no parity-dependent modifications. Left and right circular polarization modes propagate identically:

$$v_R(k, \eta) = v_L(k, \eta) \Rightarrow \Delta v = 0 \quad (\text{identically}). \quad (21)$$

No GW birefringence, no tensor chirality, no  $TB/EB$  CMB parity violation from the ECH mechanism.

### D. Explicit Verification: The Holst Term in Perturbation Theory

Expanding to second order, the Holst dual evaluates to:

$$\tilde{R}(\mathring{\Gamma}) = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\mathring{\Gamma}) = 0 \quad (\text{by the first Bianchi identity}) \quad (22)$$

This holds at every perturbation order. In particular, the cubic action for  $\zeta$  (which determines the bispectrum) receives zero contribution from the Holst term. The bispectrum is therefore identical to the standard GR result.

### E. What Would Break the Transparency

The transparency result fails if: (1) matter includes fermions with nonzero spin density; (2) the gravitational action includes kinetic terms for torsion (Poincaré gauge theory); (3) non-minimal derivative couplings between the Holst sector and matter are introduced.

### F. Implications

The perturbation-transparency result establishes a clean dichotomy:

- *Perturbation observables* ( $C_\ell^{TT}$ ,  $C_\ell^{EE}$ ,  $P_k$ , bispectrum): Identical to standard GR. No ECH modifications at any order.
- *Nonperturbative parity channels* (ALP birefringence, primordial GW chirality): The relevant tests of  $\gamma$ .

### G. Discrimination Among Bouncing Cosmologies

The matter bounce prediction  $f_{\text{NL}} = -35/8$  is the strongest discriminator: it provides  $\sigma(f_{\text{NL}}) \approx 0.7$  (Fisher-ideal) to  $\sigma(f_{\text{NL}}) \approx 1.0$  (with systematics) from SPHEREx, yielding 3–5 $\sigma$  model separation. NANOGrav model comparison:  $\gamma = 3.20 \pm 0.42$  from independent re-analysis of the 15-yr free-spectrum data (GPU MCMC, companion Paper II [2] and Paper III [45]). Bounce prediction  $\gamma = 3.0$  at  $0.48\sigma$ .

## XI. THE HYBRID DARK-ENERGY LOOPHOLE

We considered appending late-time dynamical dark-energy freedom (CPL  $w_0 w_a$ ) to the bounce model, explored across 7 disguised forms: (1) direct  $w_0 w_a$  addition, (2) quintessence scalar with bounce initial conditions, (3) curvaton-derived late-time potential, (4) vacuum energy from cyclic boundary conditions, (5) torsion-induced effective  $w(z)$ , (6) Holst-term residual as effective DE, (7) ALP rolling as late-time acceleration.

All 7 forms were rejected: adding  $w_0 w_a$  to a bounce model produces the same fit improvement as adding  $w_0 w_a$  to  $\Lambda$ CDM, with no additional theoretical content from the bounce. The loophole was explored theoretically but the  $w_0 w_a$  extension was never implemented computationally in this program (the MCMC program uses stock CAMB with  $\Delta N_{\text{eff}}$  only; see companion Paper I(b) [3] for the frozen sample inventory).

TABLE III. Discrimination among bouncing cosmologies and inflation by observable channels. “✓” denotes the mechanism produces the prediction; “×” denotes it does not; “—” denotes not applicable or not computed.

| Model                                      | $f_{\text{NL}} = -35/8$             | ALP birefringence | PTA $\gamma \approx 3$ | $w_0 w_a$ DESI          |
|--------------------------------------------|-------------------------------------|-------------------|------------------------|-------------------------|
| Matter bounce (any host; not ECH-specific) | ✓                                   | (spectator)       | ✓                      | ×                       |
| Slow-roll inflation                        | × ( $f_{\text{NL}} \approx 0.015$ ) | (spectator)       | ×                      | ×                       |
| Quintom-B                                  | ×                                   | (spectator)       | —                      | consistent <sup>†</sup> |
| Cuscuton bounce                            | × ( $f_{\text{NL}} \approx 0$ )     | (spectator)       | —                      | ×                       |
| Ekpyrotic                                  | × ( $f_{\text{NL}} \sim -5$ )       | (spectator)       | —                      | ×                       |

<sup>†</sup>Quintom-B can in principle accommodate the DESI  $w_0 w_a$  evidence; the MCMC analysis hosted in companion Paper I(b) was not extended to the  $w_0 w_a$  parameter space, so this row is reported as “consistent at the model level” rather than a posterior-preference ✓.

## XII. DISCUSSION

### A. The Inflationary Suppression Factor

The constant contribution to  $\Lambda_{\text{eff}}$  emerges from the interplay between the parity-odd spin-torsion interaction and inflationary dilution:

$$\Xi \equiv \left[ \frac{\alpha}{M} M_{\text{Pl}} \right] \times \mathcal{D}_{\text{inf}}, \quad (23)$$

a dimensionless quantity with  $\Xi \approx 10^{-123}$ , decomposed as  $10^{-2} \times \mathcal{D}_{\text{inf}}$  with  $\mathcal{D}_{\text{inf}} \sim 10^{-121}$ .

The “fine-tuning reduction from  $10^{120}$  to  $10^5$ ” is a reparameterization as sensitivity to  $N_{\text{tot}}$  (the total number of inflationary  $e$ -folds), not a resolution of the cosmological constant problem. The exponential  $\mathcal{D}_{\text{inf}} \propto e^{-3N_{\text{tot}}}$  structure of Eq. (11) makes  $N_{\text{tot}}$  the single controlling parameter analytically: the residual  $10^5$  tracks  $e^{-3\Delta N_{\text{tot}}}$  for  $\Delta N_{\text{tot}} \approx 4$   $e$ -folds, while the order-unity prefactors enter at most logarithmically.

### B. Theoretical Implications

Four routes to deriving  $\rho_\Lambda$  with  $w = -1$  from first principles were tested: (i) NJL condensate, (ii) one-loop fermion effective action, (iii) dynamical Immirzi field, (iv) parity-sensitive CMB phenomenology. All four yield clean negative results. The condensate route fails because the scalar/pseudoscalar channel is repulsive at  $\gamma = 0.274$  and subcritical. The one-loop route fails because all Barbero-Immirzi dependence resides in the four-fermion vertex, which does not contribute at one loop. The dynamical Immirzi field reduces to a standard axion-like particle with  $w = +1$  (stiff matter). The parity assessment finds no photon coupling in the minimal framework [46].

*Spectator-ALP birefringence.*—A spectator ALP with  $f_a \sim M_{\text{Pl}}$ ,  $m \sim H_0$  is consistent with the published Planck/ACT DR6  $3.6\sigma$  joint signal ( $\beta = 0.342^\circ \pm 0.094^\circ$ , Eskilt *et al.* [47]) without fine-tuning. The same  $\beta \approx 0.27^\circ$  prediction arises in standard GR with an identical ALP; it is not a distinctive ECH prediction. Full ALP MCMC parameter fitting (9,720 accepted samples,

$\hat{R} - 1 < 0.01$ ) and LiteBIRD forecast are in companion Paper I(b) [3].

## XIII. SURVIVING MECHANISM-INDEPENDENT TESTS

Despite ECH structural closure (14 barriers blocking all minimal-ECH dark-energy routes), the broader bounce-cosmology program retains two fully testable mechanism-independent predictions:

**(1) Matter-bounce  $f_{\text{NL}} = -35/8$ .**—The matter-dominated contracting phase of any matter-dominated bounce (the bounce-class observable, not specific to ECH) produces a universal non-Gaussianity  $f_{\text{NL}} = -35/8$  [1]. This value is shared by every dust-dominated contracting bounce (mechanism-independent) and is *not* a distinctive ECH prediction. It is a class-level test. The full multi-bin Fisher forecast, SPHEREx parameter sensitivity ( $\sigma(f_{\text{NL}}) \approx 0.7\text{--}1.0$ ), and anomaly-optimized multi-tracer strategy are in Paper II [2]. SPHEREx first science data  $\sim 2028$ .

**(2) Spectator-ALP birefringence  $\beta \approx 0.27^\circ$ .**—We present this as a **consistency check**, not as a prediction: the parity-odd coefficient  $\alpha/M$  is fitted, not derived from first principles, and an ALP with  $f_a \sim M_{\text{Pl}}$ ,  $m \sim H_0$  chosen to land near the observed signal is by construction inside its  $1\sigma$  band. The *quantitative* prediction is the spectral signature (frequency dependence,  $EB$  vs  $TB$  structure, scale dependence) rather than the central value of  $\beta$ . The same ALP setup arises identically in standard GR with the same parameters, so this is not a distinctive ECH prediction. LiteBIRD ( $\sigma(\beta) \approx 0.03^\circ$ , early 2030s) will either confirm a non-zero  $\beta$  at high significance or exclude the ALP explanation; either outcome is informative independent of ECH.

**Structural incompatibility.**—An open constraint is the incompatibility between the dark-energy suppression mechanism ( $N_{\text{tot}} \approx 92$   $e$ -folds required) and the  $f_{\text{NL}} = -35/8$  prediction (erased by  $N_{\text{tot}} \gg 60$ ). This is detailed in Sec. XIV D. The correct interpretation is that bounce cosmology (as a broad class) and the ECH-specific dark-energy ansatz are *independent observational programs*; SPHEREx tests the former, LiteBIRD tests a

related spectator field, and the ECH dark-energy ansatz remains a phenomenological parameterization.

## XIV. LIMITATIONS AND FUTURE DIRECTIONS

### A. Current Limitations

#### 1. Theoretical

- *Phenomenological  $\alpha/M$* : Not derived from first principles; the one-loop estimate motivates existence and order of magnitude, but the finite part depends on the  $\gamma_5$  regularization scheme (Sec. II A 2, [17]).
- *Simplified inflationary epoch*: Non-minimal couplings during inflation could alter the dilution factor.
- *Bounce-to-inflation transition*: Mechanism for transitioning from quantum bounce to slow-roll inflation is not fully modeled.

#### 2. Observational

- *Galaxy spin*: Null confirmed at the dipole level (Paper IV [20]), consistent with the  $> 100$ -orders-of-magnitude underprediction by the ECH coupling.
- *MCMC proxy*: Stock CAMB with  $\Delta N_{\text{eff}}$  is a phenomenological proxy, not a bespoke spin-torsion Boltzmann module. Full MCMC details and convergence diagnostics are in companion Paper I(b) [3].

### B. Robustness to Galaxy Spin Null Results

The galaxy spin channel is a confirmed null at the dipole level (full quantitative chirality results in Paper IV [20]), consistent with the framework (which underpredicts  $A_0$  by  $> 100$  orders of magnitude). The parity-violation case rests entirely on CMB birefringence from published Planck/ACT measurements.

### C. Discriminating Observational Channels

*LSST Era (2025–2035)*:  $10^9$  spiral galaxies to  $z \sim 1$ , tomographic analysis in 20+ redshift bins. *CMB Experiments*: LiteBIRD, CMB-S4, and future concepts (PICO, CMB-HD). *Bounce cosmology beyond ECH*: Papanikolaou *et al.* [44] showed that asymmetric matter bounces can produce asteroid-mass PBHs as dark matter candidates with induced GWs detectable by LISA and Einstein Telescope.

### D. Structural Tension: Dark Energy vs. Bounce $f_{\text{NL}}$

The dark-energy suppression mechanism requires  $N_{\text{tot}} \approx 92$  post-bounce  $e$ -folds; the matter-bounce  $f_{\text{NL}} = -35/8$  would be erased by  $N_{\text{tot}} \gg 60$ . These are structurally incompatible within minimal ECH. DESI DR2 evidence for equation-of-state crossing at  $3.1\text{--}4.2\sigma$  [7] lends empirical support to quintom scenarios that can unify bounce and dark energy through mechanisms outside the ECH minimal-route catalog.

### E. Structural Closure

The 14 structural constraints catalog the complete set of minimal-ECH dark-energy mechanisms and close all of them. The surviving science case rests on the matter-bounce  $f_{\text{NL}}$  prediction, which is compatible with the ECH framework but not derived from it (Sec. XIII).

## XV. CONCLUSIONS

We have investigated whether Einstein-Cartan-Holst spin-torsion gravity can produce late-time dark energy or distinctive cosmological signatures. The answer is a structural no-go: the 14 independent constraints (Table II) close every minimal-ECH dark-energy route.

*Central result: perturbation transparency.*—For canonical scalar field matter, the Holst sector decouples completely from all scalar and tensor perturbation observables (Sec. X). Torsion vanishes at all perturbation orders; the Holst dual contraction vanishes identically by the first Bianchi identity. This is a positive structural result: it identifies the nonperturbative parity-violating channels (ALP birefringence, primordial GWs) as the relevant tests of  $\gamma$ .

*The 14 barriers.*—Systematic analysis across 7 foundations (A–G) and 7 observational branches established 14 independent structural constraints on the minimal ECH dark-energy parameter space (Sec. IX). These constraints are ECH-specific: other bouncing cosmologies (notably quintom scenarios) can in principle unify bounce with late-time dark energy through mechanisms outside the ECH minimal-route catalog.

*Surviving tests.*—Two mechanism-independent predictions of the broader bounce/ALP landscape survive ECH structural closure and are testable:

1.  $f_{\text{NL}} = -35/8$  (matter-bounce class): SPHEREx tests at  $3\text{--}5\sigma$  realistic significance by  $\sim 2028$ , discriminating from inflation ( $f_{\text{NL}} \approx 0.015$ ) and the Cuscuton bounce ( $f_{\text{NL}} \approx 0$ ).
2. Spectator-ALP birefringence  $\beta \approx 0.27^\circ$ : LiteBIRD tests at  $\sim 9\sigma$  in the early 2030s (confirm or cleanly exclude).



Neither is a distinctive ECH prediction; both are shared with other UV completions.

*Known limitations.*—This work does not derive the IR effective vacuum term from first principles:  $w = -1$  is assumed, not derived; the birefringence prediction lacks a derived photon-torsion coupling;  $\alpha/M$  is a phenomenological parameter; and the MCMC uses stock CAMB (not a bespoke torsion-modified Boltzmann code). Full MCMC diagnostics, ALP parameter fitting, and NaMaster pipeline validation are in companion Paper I(b) [3].

*Forward.*—The program continues with direct tests of the bounce framework: SPHEREx  $f_{\text{NL}}$  forecast (Paper II [2]), multi-survey anomaly catalog (Paper III [45]), and galaxy chirality catalog (Paper IV [20]).

### Data and Code Availability

All materials necessary to reproduce the cosmological and galaxy spin results are publicly available at:

<https://github.com/Hubify-Projects/bigbounce/tree/v2.3.0/reproducibility>

(pinned to tag v2.3.0). The repository includes Cobaya YAML configurations, galaxy spin pipeline code, and the implementation map (IMPLEMENTATION\_MAP.md). MCMC chains and convergence diagnostics are in companion Paper I(b) [3].

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### Appendix A: Complete Parameter Summary

### Appendix B: Dimensional Analysis

The parity-odd operator (Eq. 6) has off-shell mass dimension +1, not the +4 required for a Lagrangian density:

$$[\alpha/M] = -1, \quad [\varepsilon^{\mu\nu\rho\sigma} e_\mu^I e_\nu^J \mathcal{F}_{IJ\rho\sigma}] = +2 \implies [\mathcal{L}_{\text{odd}}] = +1. \quad (\text{B1})$$

The missing three powers of mass arise through on-shell evaluation at Planck-scale bounce densities:  $K \sim M_{\text{Pl}}$  and  $R \sim M_{\text{Pl}}^2$ , giving  $\rho_\Lambda^{\text{bounce}} \sim (\alpha/M) M_{\text{Pl}}^3 = [(\alpha/M) M_{\text{Pl}}] M_{\text{Pl}}^4 \sim 10^{-2} M_{\text{Pl}}^4$ . Inflationary dilution ( $\mathcal{D}_{\text{inf}} \sim e^{-3N_{\text{tot}}}$ ) then yields  $\rho_\Lambda = \Xi M_{\text{Pl}}^4$  with  $[\rho_\Lambda] = +4$ . This is a *scaling ansatz*, not a derivation from a renormalizable off-shell EFT. No quantitative conclusion in this paper depends on resolving this gap: the 14 structural barriers close the bounce-to-dark-energy route on independent grounds.

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TABLE IV. Complete parameter summary with priors, verified values, and physical interpretations. MCMC posterior values are from companion Paper I(b) [3].

| Parameter                                                  | Description            | Prior             | Verified Value                    | Notes                        |
|------------------------------------------------------------|------------------------|-------------------|-----------------------------------|------------------------------|
| <i>Fundamental theory parameters</i>                       |                        |                   |                                   |                              |
| $\gamma$                                                   | Barbero-Immirzi        | Fixed: 0.274      | $0.274 \pm 0.020$                 | LQG area spectrum            |
| $\alpha/M$                                                 | Parity-odd coefficient | Log-flat          | $\sim 10^{-21} \text{ GeV}^{-1}$  | One-loop motivated           |
| $N_{\text{tot}}$                                           | Total e-folds          | [60, 120] flat    | $\approx 92$ (fitted)             | Controls $\Xi$               |
| <i>Cosmological parameters (MCMC proxy from companion)</i> |                        |                   |                                   |                              |
| $H_0$                                                      | Hubble constant        | From Planck prior | $67.68 \pm 1.06 \text{ km/s/Mpc}$ | Recovers $\Lambda\text{CDM}$ |
| $\Delta N_{\text{eff}}$                                    | Radiation proxy        | $[-3, 3]$ flat    | $-0.020 \pm 0.169$ (full-tension) | Consistent with 0            |
| $\sigma_8$                                                 | Clustering amplitude   | Derived           | $0.803 \pm 0.008$                 |                              |
| $\Omega_m$                                                 | Matter density         | Derived           | $0.308 \pm 0.005$                 |                              |
| <i>Observational channel parameters</i>                    |                        |                   |                                   |                              |
| $\beta$                                                    | ALP birefringence      | —                 | $0.27^\circ$ (midpoint)           | Spectator ALP, not ECH       |
| $f_{\text{NL}}$                                            | Non-Gaussianity        | —                 | $-35/8 = -4.375$                  | Matter bounce class          |
| $\gamma_{\text{PTA}}$                                      | PTA spectral index     | —                 | $3.20 \pm 0.42$ (GPU MCMC)        | Bounce at $0.48\sigma$       |

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